

Sri Lanka Institute of Information Technology



IT2140 | Database Design and Development

Assignment Part 01 - Requirement Gathering

Zentra Mart — Web-Based Marketplace Platform

Group ID: 2026-Y2-S1-KU-19

Group Members

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1. System Purpose

Zentra Mart is a web-based marketplace platform that connects buyers and sellers on a single system. The platform allows sellers to list and manage products, and allows buyers to search, purchase, and track orders. Zentra Mart is intended to replace the fragmented tools (social media posts, direct messages, spreadsheets) that individual sellers currently rely on, and to give both sellers and buyers one connected, trustworthy environment for listing, discovery, purchase, and after-sales support.

The platform also supports order fulfilment through a delivery workflow, secure payment processing through an external payment gateway, automated notifications for key events, and administrative oversight to keep the marketplace safe and well-moderated.

2. Target Users

User / Actor	Description
Buyer	A registered customer who searches for products, manages a shopping cart, completes purchases, and tracks orders.
Seller	A registered vendor who lists products, manages inventory, and fulfils incoming orders.
Administrator	Platform staff who manage user accounts, moderate product listings, and review customer complaints.
Delivery Partner	An individual or courier responsible for accepting, transporting, and confirming the delivery of orders.
Payment Gateway (external)	A third-party service that verifies and processes payment transactions on behalf of the platform.
Notification Service (external)	A supporting service that sends email/SMS/in-app notifications for events such as order confirmation and status updates.

3. Functional Requirements

3.1 Account Management

- The system shall allow a visitor to register an account and select a role (Buyer or Seller).
- The system shall verify new accounts using a one-time password (OTP) sent via the Notification Service.
- The system shall allow registered users to log in, edit their profile, and reset a forgotten password.

3.2 Product Listing Management

- The system shall allow a seller to create a product listing with a name, description, price, stock quantity, category, and one or more images.
- The system shall allow a seller to edit or archive an existing listing.

3.3 Search and Browsing

- The system shall allow a buyer or guest to search products by keyword and filter results by category and price range.
- The system shall allow a buyer to sort results by price, newest, or rating.

3.4 Shopping Cart and Checkout

- The system shall allow a buyer to add products to a cart and adjust quantities.
- The system shall allow a buyer to apply a discount code during checkout.
- The system shall reserve stock and invoke the Payment Gateway to verify payment before confirming an order.
- The system shall create an order record and request an order confirmation notification once payment is verified.

3.5 Order and Delivery Management

- The system shall allow a seller to view incoming orders and update the order status (e.g., Shipped).
- The system shall assign a delivery partner to a confirmed order and allow the delivery partner to update delivery status through to confirmation.
- The system shall allow a buyer to track the current status and history of their orders.

3.6 Administration

- The system shall allow an administrator to manage user accounts, including suspending or reactivating an account.
- The system shall allow an administrator to moderate flagged or reported product listings.
- The system shall allow an administrator to review and resolve customer complaints.

4. Non-Functional Requirements

Quality	Requirement
Performance	Search results shall return within 3 seconds; checkout shall complete within 10 seconds of payment confirmation.
Security	Passwords shall be stored using a secure hash; sessions shall be token-based; access shall be restricted by role.
Usability	The interface shall be mobile-first and responsive, with a simple, guided checkout flow.
Scalability	The system architecture shall support a growing number of sellers, products, and orders without redesign.
Reliability	The system shall target 99% uptime; order and payment records shall be backed up regularly.
Compatibility	The system shall function correctly on current versions of Chrome, Edge, and Firefox.

5. Constraints

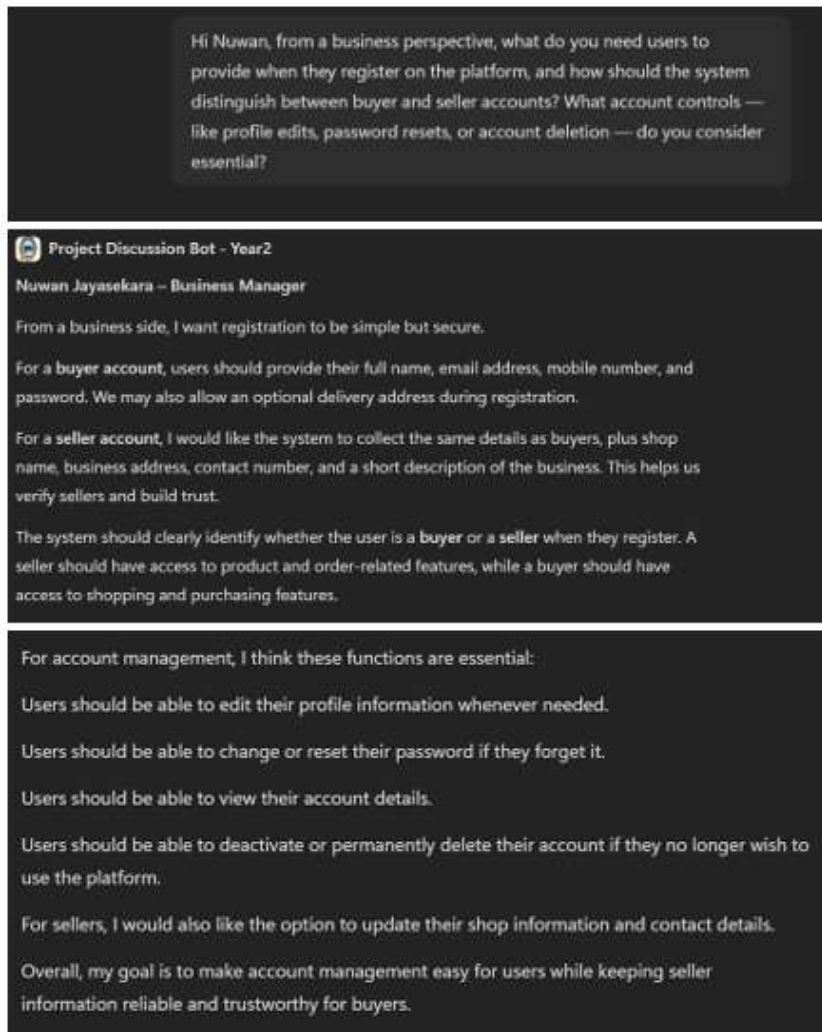
- One product stock unit corresponds to one active reservation; stock cannot go negative.
- A confirmed order must have exactly one associated payment record.
- Payment must be verified by the Payment Gateway before an order is confirmed and a delivery is assigned.
- Only an administrator may suspend a user account or remove a product listing.
- A discount code may only be applied to an order if it is valid and unexpired at the time of checkout.
- The initial release supports a single currency and the English language only.
- Payment processing is simulated for this assignment and is not connected to a live payment gateway.

6. Evidence of Requirement Gathering (Bot Interaction)

Requirements for this system were gathered through interviews conducted using the module's Project Discussion Bot. Each interview began by asking how the business currently operates and what problems staff encounter, before moving to specific feature requests.

Interview Screenshots:

Dialogue Stream 1: User Registration & Account Management (Nuwan Jayasekara , Business Manager)



Dialogue Stream 2: Product Listing Management (Chamara Wijesinghe, Seller Operations Executive)

Hi Chamara, as the Seller Operations Executive, what details do sellers need to provide when creating a new product listing? What tools do they need to update stock, adjust pricing, or mark an item as sold or unavailable?

Chamara Wijesinghe – Seller Operations Executive

For a new product listing, I want sellers to provide the product name, category, description, price, available stock quantity, and clear product images. It is also useful to include item condition, brand, and delivery information so buyers can make good decisions.

After creating a listing, sellers should be able to easily edit any product details. They should be able to update stock levels when items are added or sold, change prices when needed, and update product descriptions or images.

I also need a simple option for sellers to mark products as **Sold Out**, **Unavailable**, or **Active**. This helps prevent buyers from ordering items that cannot be supplied. Sellers should also be able to view all their listings in one place and quickly see which products are active, low in stock, or sold out.

From our operations side, it would be helpful if the system warns sellers when stock is running low so they can update inventory before customers are affected.

Dialogue Stream 3: Search & Browse / Product Catalog (Tharushi Perera, Customer Experience Officer)

Hi Tharushi, from the customer's side, how do buyers typically search for products, and what filtering or sorting options make that easier? What information should a product detail page show before a customer decides to buy?



Project Discussion Bot - Year2

Tharushi Perera – Customer Experience Officer

From a customer's point of view, searching should be quick and simple. Most buyers search using product names or keywords. They also like to browse by category.

To make searching easier, I think the system should allow filtering by category, price range, product condition, seller, and product availability. Buyers also like sorting options such as price low to high, price high to low, newest products, and highest-rated products.

Before buying, customers need enough information on the product page to make a decision. I would like the product detail page to show the product name, clear images, price, description, category, available stock, seller name, seller rating, delivery information, and customer reviews.

Customers should also be able to see whether the item is in stock, how many ratings it has received, and the average rating. This helps build trust and reduces complaints after purchase.

When buyers have questions, there should be  easy way to contact the seller directly from the product page.

Dialogue Stream 4: Shopping Cart & Checkout (Tharushi Perera, Customer Experience Officer)

Hi Tharushi, walk me through the ideal cart and checkout experience for a customer — adding or removing items, entering a shipping address, and completing a simulated payment. What would make that process feel smooth and trustworthy?



Project Discussion Bot - Year2

Tharushi Perera – Customer Experience Officer

From the customer side, I want the buying process to be very easy.

When a buyer finds a product, they should be able to click "Add to Cart". In the cart, they should be able to see all selected items, change quantities, remove products, and view the total cost before checkout.

During checkout, the buyer should enter or select a shipping address. If they have ordered before, it would be helpful if the system remembers saved addresses.

Before placing the order, the customer should see an order summary with the product details, quantities, delivery address, and total amount. This allows them to check everything before confirming.

For this project, a simulated payment process is fine. The system should clearly show that the payment was successful and generate an order confirmation with an order number.

To make the process feel trustworthy, customers should always know what step they are on. Clear prices, no hidden charges, confirmation messages, and order receipts are very important. After checkout, they should receive a notification confirming that the order was placed successfully.

As a customer, I also like having an easy way to go back and edit my cart before making the final purchase. This reduces mistakes and improves confidence when ordering.

Dialogue Stream 5: Order Management & Tracking (Asanka Ranasinghe, Logistics & Order Coordination Officer)

Hi Asanka, once an order is placed, how should it move through fulfillment — from pending to shipped to delivered? What do sellers need to update order status, and what should buyers be able to see when tracking their own orders?



Project Discussion Bot - Year2

Asanka Ranasinghe – Logistics & Order Coordination Officer

Once a customer places an order, I would like the order to move through a simple process.

First, the order should be marked as **Pending**. This means the order has been received and is waiting for the seller to prepare it.

After the seller packs and dispatches the item, the status should change to **Shipped**. At this stage, the item is on its way to the customer.

When the customer receives the item successfully, the seller can update the status to **Delivered**. This completes the order process.

For sellers, the system should provide an easy way to view incoming orders and update the order status. Sellers should also be able to add a tracking number or delivery note if available.

For buyers, I want them to see:

- Order number
- Order date
- Products ordered
- Current order status (Pending, Shipped, Delivered)
- Delivery address
- Tracking number (if provided)
- Order history

It is also important that buyers receive notifications whenever the order status changes. For example, they should be informed when the order is shipped and when it is marked as delivered.



Dialogue Stream 6: Admin Management Panel (Dilini Karunaratne, Marketplace Administrator)

Hi Dilini, as the Marketplace Administrator, what tools do you need to approve or suspend seller accounts, manage product categories, and remove listings that break platform rules? What platform-wide statistics or dispute-resolution tools would help you most?



Project Discussion Bot - Year2

Dilini Karunaratne – Marketplace Administrator

As the administrator, I need a dashboard where I can review seller accounts before they start selling. When a seller registers, I should be able to approve or reject the account. If there are complaints or rule violations, I need the ability to suspend or reactivate seller accounts.

For product categories, I need tools to create new categories, edit existing categories, and remove unused categories. This helps keep the marketplace organised and makes products easier to find.

For listings, I need to view all products on the platform. If a listing contains inappropriate content, misleading information, prohibited items, or spam, I should be able to remove it immediately. It would also be useful to receive reports submitted by users so I can investigate problematic listings.

For platform statistics, I would like to see:

- Total number of buyers and sellers
- Total products listed
- Total orders placed
- Most popular product categories
- Active vs suspended sellers
- Recent sales activity

For dispute resolution, I need a system where buyers and sellers can submit complaints. I should be able to view dispute details, review messages related to the issue, record decisions, and mark disputes as resolved. Keeping a history of disputes is also important for future reference.

These tools will help ensure the marketplace remains safe, trustworthy, and well-managed for all users.